

TECHNOLOGICAL UNIVERSITY (THANLYIN)
DEPARTMENT OF INFORMATION TECHNOLOGY ENGINEERING

**OBSTACLE DETECTION FOR RASPBERRY PI BASED
RESTAURANT SERVING ROBOT**

BY
MA INKYINN THAW THAW
VI IT - 2

GRADUATION THESIS

OCTOBER, 2025
THANLYIN

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**A THESIS
SUBMITTED TO THE DEPARTMENT OF
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IN PARTIAL FULFILMENT OF THE REQUIREMENTS
FOR THE DEGREE OF BACHELOR OF ENGINEERING
(INFORMATION TECHNOLOGY ENGINEERING)**

OCTOBER, 2025

THANLYIN

TECHNOLOGICAL UNIVERSITY (THANLYIN)
DEPARTMENT OF INFORMATION TECHNOLOGY ENGINEERING

We certify that we have examined, and recommend to the University Steering Committee for Bachelor of engineering Graduation thesis entitled: **“OBSTACLE DETECTION FOR RASPBERRY PI BASED RESTAURANT SERVICING ROBOT”** submitted by **MA INKYINN THAW THAW**, Roll No. **VI-IT 2 (November, 2024)** to the Department of Information Technology Engineering as the requirement for the Bachelor of Engineering degree **B.E (IT)**.

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ABSTRACT

This thesis presents the design and implementation of an obstacle detection system integrated with line-following functionality for a restaurant serving robot using a Raspberry Pi. The robot autonomously navigates by following a black line on the floor using infrared sensors that detect line contrast. To ensure safe operation in dynamic environments, a Time-of-Flight (TOF) sensor is used to detect obstacles ahead. When an obstacle is identified within a defined range, the robot halts its movement and simultaneously issues a voice alert through connected speakers to notify nearby users. The Raspberry Pi serves as the central processing unit, continuously collecting sensor data, processing information in real time, and executing motor control commands alongside alert activation. The software, developed in Python, employs threshold-based decision-making to maintain accurate line tracking while responding promptly to obstacles. This straightforward yet effective method avoids the complexity of rerouting algorithms, making the system well-suited for structured indoor environments such as restaurants. Experimental testing demonstrates that the robot reliably follows its designated path and detects obstacles with accuracy, stopping and alerting as intended. The voice alert mechanism enhances user interaction and safety, while the modular system design allows for potential future improvements, such as rerouting functions or integration of more advanced sensors. Overall, this work contributes to the advancement of service robots in hospitality by combining reliable navigation with efficient obstacle detection and interactive voice feedback.

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